ICT 4442 — Deep Learning Project
Interim Report (Part B)

Team No. and Names: CalmGesture (Team No: 420)
Members: Sangini Singh (230911020); Tanvi Gupta (230911122); Aditya Nigam (230911366); Shubh Rastogi (230953258)
Title of the Project: CalmGesture: A Gesture-to-Sound Sensory Regulation Tool for Autism Support Using Deep Learning
GitHub Repository Link: https://github.com/aditya1nigam8/calmgesture

1. Literature Review

Ten papers were reviewed across four themes: autism-related repetitive-movement analysis, gesture-to-music interaction, landmark-based gesture classification, and sequence models for gestures. Details below are taken from each paper's abstract.

Paper (Author, Year)	Method	Dataset	Key Result	Relevance to Project
[1] Rajagopalan, Dhall & Goecke, 2013	Bag-of-words action recognition baseline	SSBD: 75 in-the-wild videos; arm flapping, head banging, spinning	First public dataset of self-stimulatory behaviours; baseline recognition reported (no headline figure in abstract)	Shows camera-based analysis of repetitive movement is feasible. Related work only; not used for training.
[2] Mohammadian Rad et al., 2015	CNN feature learning + SVM on accelerometer signals; transfer learning	6 autistic subjects, wrist/torso 3-axis accelerometers, two studies 3 years apart	Mean F1 (Study 1 / 2): handcrafted 0.54 / 0.40; CNN 0.74 / 0.35; transfer-learning CNN 0.78 / 0.53	Learned features beat handcrafted ones for repetitive-movement detection. Wearable modality differs from our camera input.
[3] Mohammadian Rad et al., 2017	CNN features, LSTM temporal modelling, transfer learning, LSTM ensembles on IMU data	Simulated data + two real stereotypical-movement datasets	Feature learning outperformed handcrafted; LSTM ensembles gave more accurate, stable detectors (no figures in abstract)	Temporal models help on repetitive movement; supports including a GRU in our comparison.
[4] Zhang et al., 2022	RNN classifying EEG of children during gesture-interactive-robot music sessions	10 autistic + 10 healthy children (3-12 y), EEG	Accuracy 72-94% (mean about 85%); RNN beat CNN (57-61%), LSTM (64-74%), SVM (53-75%)	Closest precedent linking gesture interaction, music and autistic children. Note: classification was on EEG, not gestures.
[5] Jeyaraj et al., 2025 (Gesture2Music)	Causal temporal CNN on body/hand landmarks predicting note-level events	Custom: 5 volunteers, 21 gesture-note classes, 3,150 clips	97.9% pitch accuracy; 25-30 ms inference, 60-70 ms loop latency; GRU baseline 94.26%, LSTM 94.7%	Shows landmark-to-sound is feasible in real time and gives a latency budget; includes a GRU baseline.
[6] Singh et al., 2024	MediaPipe 21 landmarks + CNN (3 conv, 3 pool, 2 FC)	ASL Alphabet: 87,000+ samples, 29 classes	99.12% accuracy, real time	Direct precedent for our MediaPipe-landmark + CNN pipeline.
[7] Gil-Martin et al., 2025	MediaPipe landmarks + recurrent nets; landmark formats, normalisation and sequence lengths compared	IPN Hand, subject-wise cross-validation	Best 84.66 +/- 1.09% (wrist of first frame as reference); 4 landmarks alone still 81.46%	Informs our normalisation choice; shows cross-user accuracy is far lower than within-user figures.
[8] Pooja et al., 2025	GRU-LSTM on hand-gesture sequences	ASL alphabet (dataset not named in abstract)	99.39% accuracy; CNN 96.58%, DNN 97.65%, SVM 98.09%, RNN 98.72%	Supports recurrent models for gesture sequences (our GRU).
[9] Zhang, Liu & Wu, 2019	2D CNN + Temporal Segment Network with Temporal Shift and SE modules on mobile backbones	Kinetics-400; Jester as case study	MnasNet+TSM+SE: 93.7% on Jester	Establishes Jester as a benchmark and shows mobile/real-time feasibility.
[10] Linardakis et al., 2025	Survey of 137 papers (2018-2025)	Multiple benchmark datasets	Persistent challenges: real-world robustness, occlusion, cross-user generalisation, efficiency	Frames our four-family comparison; flags cross-user generalisation as a risk.

Taken together, the literature supports a MediaPipe-landmark front end feeding lightweight classifiers [6, 7], suggests recurrent or temporal models should help on repetitive motion [3, 5, 8], and warns that cross-user generalisation is the main weakness of small gesture systems [7, 10]. Papers [1]–[4] show that camera- and sensor-based analysis of autism-related movement and music interaction is an active area, but none of them maps recognised hand movements to sound feedback, which is the gap CalmGesture targets.

2. Dataset Acquisition & Preprocessing

Datasets. (i) A subset of the 20BN-Jester gesture videos (rolling hand, shaking hand, drumming fingers, swiping and similar classes) and (ii) a small custom webcam dataset recorded by the team (rolling, shaking, waving, finger tapping, hand open/close). SSBD is treated as related work only.

Pipeline (preprocess.py in the repository):

- Landmark extraction: MediaPipe Hands, one hand, 21 landmarks × (x, y, z) per frame, up to 60 frames per clip. Frames with no detection reuse the last valid landmarks so every clip is a continuous sequence.
- Normalisation: landmarks are made wrist-relative and divided by the wrist–middle-finger-base distance, making them invariant to hand position and distance from the camera.
- Fixed length: each clip is linearly resampled to 30 frames, giving a (30, 63) array saved as .npy.
- Model inputs: the MLP receives the per-landmark mean and standard deviation over time (126 features, standardised). The CNN receives a 2 × 21 × 3 "pose image" (channel 0 = mean pose, channel 1 = temporal standard deviation), standardised per channel. The 1D-CNN and GRU will consume the full (30, 63) sequence.
- Split: stratified 70% train / 10% validation / 20% test with a fixed seed (42), shared by all models.

Data status. The pipeline code is complete, but the Jester subset and the custom recordings have not yet been processed through it. The results in Section 3 were therefore produced on 200 procedurally generated synthetic landmark sequences (5 classes × 40 clips) in the same (30, 63) format, solely to validate the pipeline end to end. [TEAM: replace with real-data processing details and clip counts once available.]

3. Models Implemented So Far

Model	Owner (Member)	Status	Preliminary Metric	Notes
MLP	Sangini Singh	Implemented; run on synthetic pipeline-check data	Test acc. 95.1%, macro-F1 94.9% (41 test clips)	scikit-learn MLP, hidden layers (64, 32), ReLU, early stopping; 126-d standardised mean/std features.
CNN	Tanvi Gupta	Implemented; run on synthetic pipeline-check data	Test acc. 100%, macro-F1 100% (41 test clips); 1.5 s training, 0.06 ms/clip inference	From-scratch NumPy 2D CNN (2 conv + 2 dense) on the 2×21×3 pose image. PyTorch port and Gesture-Symphony-based architecture still to do.
1D-CNN	Aditya Nigam	Not started	—	Planned: 1D convolution over the (30, 63) landmark sequence. Target 10 Oct 2026.
GRU	Shubh Rastogi	Not started	—	Planned: GRU over the (30, 63) sequence. Target 10 Oct 2026.

Reading these numbers. The synthetic classes are cleanly separable, so near-ceiling scores confirm that the code works but say nothing about accuracy on real gestures; they must not be quoted as project results. One observation is likely to carry over: a first CNN variant that used only the mean pose reached just 41.5% on the same split, because the gesture classes differ in motion rather than static shape. Adding the temporal standard-deviation channel fixed this, which supports testing explicit temporal models (1D-CNN, GRU) on real data. All four models will be re-run on the real Jester subset and custom data under the same split and metrics (accuracy, macro-F1, confusion matrix, inference latency).

4. Individual Contribution Log (till 2nd October 2026)

Member Name	Reg. No.	Task(s) Completed	Signature
Sangini Singh	230911020	Built the MLP baseline and evaluated it on the synthetic pipeline-check data with the shared split. Re-running on real data is listed as remaining.	
Tanvi Gupta	230911122	Built the NumPy 2D CNN and evaluated it on the synthetic data. The entry notes the finding that a temporal-std channel was needed. The PyTorch port is listed as remaining.	
Aditya Nigam	230911366	Shared pipeline, common split and metrics, and the report draft. The 1D-CNN is listed as not started, due 10 Oct 2026.	
Shubh Rastogi	230953258	Shared work, with the GRU listed as not started and due 10 Oct 2026.	

5. Risk / Plan for Remaining Work

Remaining models: 1D-CNN (Aditya Nigam) and GRU (Shubh Rastogi), then comparison of all four on real data and integration of the best model into the real-time gesture-to-sound application.

Risk / Challenge	Mitigation	Owner / Target
Real data not yet processed; reported metrics are pipeline checks only	Run preprocess.py on the Jester subset and custom clips first; re-run MLP and CNN on the real split before any comparison	All members; first task before 10 Oct
Jester is large and needs registration	Use only 4–5 selected classes and cap clips per class; keep the subset list in the repository	All members
Jester-to-custom domain gap; small custom set; cross-user generalisation [7, 10]	Subject-wise splits for the custom data; report Jester and custom results separately; light augmentation (noise, time-warp)	All members
MediaPipe misses hands in fast motion or poor light	Log per-clip detection rate; drop clips below a threshold; record custom data in good light	All members
CNN is currently NumPy-only	Port to PyTorch in a GPU-enabled environment and adapt the Gesture-Symphony architecture	Tanvi Gupta; before 10 Oct
Real-time latency in the final app	Measure per-model inference time; target the 25–30 ms range reported in [5]	Final integration, 31 Oct
Non-clinical scope	State clearly that the system is an exploratory prototype making no diagnostic or therapeutic claims	All members

Timeline (from the synopsis work plan): 1D-CNN and GRU trained and all four models compared by 10 Oct 2026; best model integrated into the real-time application, error analysis completed and Final Report submitted by 31 Oct 2026.

References

[1] S. S. Rajagopalan, A. Dhall, R. Goecke, "Self-Stimulatory Behaviours in the Wild for Autism Diagnosis," ICCV Workshops, 2013.

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[5] R. Jeyaraj, B. Subramanian, K. Gangadharan, A. Paul, "Gesture2Music: A Low-Latency Real-Time Framework for Continuous Gesture-Driven Music Generation," arXiv:2511.00793.

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[7] M. Gil-Martin et al., "Hand Gesture Recognition Using MediaPipe Landmarks and Deep Learning Networks," SciTePress, 2025, paper 130535.

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